American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Preparation Guide

Faculty: Mejbah Ahammad | **Email:** ahammadmejbah@aiub.edu

Mid-Term Examination Preparation Guide | **Date: July 24, 2026**

1 Recommended Study Topics

1.1 Module 1: Networking Fundamentals

Students should thoroughly understand:

- TCP/IP Protocol Suite
- Layer Responsibilities
- Encapsulation and Decapsulation
- Repeater
- Hub
- Bridge
- Switch
- Router
- Collision Domain
- Broadcast Domain
- Ethernet Standards
- WLAN Standards

Students should be able to explain the purpose of each networking device and identify the layer in which it operates.

1.2 Module 2: Application Layer

Prepare the following concepts:

- Protocol Fundamentals
- HTTP
- DNS
- FTP
- SMTP
- Push Protocol
- Pull Protocol
- Recursive DNS Resolution
- Iterative DNS Resolution

Students should understand the communication process rather than memorizing protocol definitions.

1.3 Module 3: Data Link Layer

Study the following topics carefully:

- Logical Link Control (LLC)
- Medium Access Control (MAC)
- Multiple Access Techniques
- FDMA
- TDMA
- CDMA
- Reservation
- Polling
- Token Passing

Students should understand when each technique is used and its advantages and limitations.

1.4 Module 4: Random Access Protocols

Focus on understanding:

- Pure ALOHA
- Slotted ALOHA
- CSMA
- 1-Persistent CSMA
- Non-Persistent CSMA
- p-Persistent CSMA
- CSMA/CD
- CSMA/CA

Pay particular attention to collision handling, efficiency, throughput, and channel utilization.

1.5 Module 5: Routing

Students should understand:

- Routing Fundamentals
- Routing Table
- Static Routing
- Dynamic Routing
- Routing Information Protocol (RIP)
- Hop Count
- Route Advertisement
- Route Update
- Network Failure Recovery

Practice updating routing tables using different network topologies.

1.6 Module 6: DHCP and ARP

Study both protocols conceptually.

DHCP

- Purpose
- DORA Process
- Lease Time
- DHCP Messages

ARP

- Purpose
- ARP Request
- ARP Reply
- ARP Cache
- Cache Timeout

Students should understand how logical addresses are translated into physical addresses.

2 Laboratory Preparation

Students should practice solving numerical problems involving:

IP Addressing

- Binary to Decimal Conversion
- Decimal to Binary Conversion
- Address Classes
- Network ID
- Host ID
- Broadcast Address
- Prefix Notation

FLSM

Students should confidently determine:

- Number of Subnets
- Block Size
- Network Address
- Broadcast Address
- First Host Address
- Last Host Address
- Number of Usable Hosts

VLSM

Practice:

- Sorting Host Requirements
- Selecting Appropriate Prefix Length
- Allocating Address Blocks
- Preventing Address Overlap
- Calculating Remaining Address Space

3 Problem Solving Practice

Students are encouraged to practice original networking problems involving:

- Enterprise Network Design
- IPv4 Address Planning
- Network Expansion
- Subnet Allocation
- Routing Table Updates
- Multiple Access Communication
- DNS Resolution
- DHCP Operation
- ARP Communication
- Collision Domain Analysis
- Broadcast Domain Analysis

4 Important Concept Comparisons

Prepare comparison tables for:

- Hub vs Switch
- Switch vs Router
- Static Routing vs Dynamic Routing
- FLSM vs VLSM
- DHCP vs BOOTP
- ARP vs RARP
- Recursive DNS vs Iterative DNS
- Pure ALOHA vs Slotted ALOHA
- CSMA/CD vs CSMA/CA
- TCP/IP Model vs OSI Model

5 Recommended Diagrams

Students should practice drawing and explaining:

- TCP/IP Architecture
- Encapsulation Process
- DHCP Message Exchange
- ARP Communication Flow
- Routing Topology
- TDMA Frame Structure
- CDMA Encoding
- CSMA Operation
- Collision Domain Illustration
- Broadcast Domain Illustration

6 Numerical Preparation

Students should regularly practice calculations involving:

- Binary Conversion
- Prefix Length
- Subnet Masks
- Block Size
- Host Calculation
- Network Address
- Broadcast Address
- FLSM Design
- VLSM Design
- Routing Metrics

Show all intermediate steps while solving numerical problems.

7 Conceptual Preparation

Rather than memorizing definitions, students should always ask:

1. What is the concept?
2. Why is it required?
3. How does it work?
4. Where is it applied?
5. What are its advantages?
6. What are its limitations?

8 Examination Strategy

1. Read every question carefully before answering.
2. Identify the key networking concept involved.
3. Draw diagrams whenever appropriate.
4. Show all calculation steps for numerical problems.
5. Present answers in a clear and organized manner.
6. Use proper networking terminology.
7. Verify all subnetting and routing calculations before submission.
8. Allocate time wisely to complete all required questions.